Bonus 1: A scientific Ohm-law figure

Physics Programming, Week 1 — Optional extra credit

Optional bonus: make a scientific figure communicate

This bonus is optional, is not a prerequisite for later work, and is not examined. A completed bonus is worth **one-third of a final-grade point**. Before you start, read this week's lecture together with the self-contained reading below; the optional *Going further: Lissajous curves* companion is not needed for this bonus.

After 10–15 minutes of self-contained reading, we fit a supplied Ohm-law data set and turn it into one clear figure. Here voltage is the horizontal coordinate and current the vertical coordinate, so the model slope is conductance in S, not resistance in Ω. We identify the measured-current series and the fitted-current series before changing their appearance.

We add the prescribed title, labels with units, legend entries, observation markers, model line, frame, and canvas size, then export a PDF. This bonus is graded **by hand** against the checklist below (not by CodeRunner). Changing a label or line style must not alter a measurement.

Scope notice

This bonus is optional, is not a prerequisite for later work, is not examined, and is worth **1/3 point on the final course grade**.

Self-contained reading (10–15 minutes)

Physics box: an ohmic conductor

Voltage drives a current through a conductor. In the ohmic regime the current is proportional to the applied voltage, so $V = RI$, equivalently $I = gV$. Voltage V is in volts, current I in amperes, resistance R in ohms (Ω), and conductance $g = 1/R$ in siemens (S). We take the positive current in a chosen reference direction through the conductor; reversing that direction flips the sign of both I and the fitted intercept. The relation assumes a fixed conductor state, a weak field, and linear response, so R stays constant over the measurement. The ideal check is $V = 0 \Rightarrow I = 0$. Heating changes R, strong fields and non-ohmic devices bend the curve, and contacts create offsets, so a straight line fitted here holds only over the measured range. Scattering establishes steady drift and helps set the resistance, but rapid collisions alone do not guarantee linearity.

With current on the vertical axis and voltage horizontal, a fitted line $I = gV + I_0$ has slope g equal to conductance in S and intercept I_0 in A. The ideal check is $V = 0 \Rightarrow I = 0$.

Task: fit the data and draw one figure

Work with the supplied `ohm_law.csv` in the course data folder (columns `voltage_V` in V and `current_A` in A).

1. Using the same centered-sum recipe as the Week 1 lecture, fit $I = gV + I_0$ and keep the conductance g (in S), the intercept I_0 (in A), and the fitted current at every supplied voltage.

2. Draw one figure with the prescribed title `Ohm-law measurements and fitted model`; horizontal label `Voltage (V)`; vertical label `Current (A)`; measured

points as markers labelled `measurements`; the fitted line labelled `least-squares model`; a boxed frame; and canvas size $(800, 500)$ pixels. Both series share one vertical axis, current in A.

3. Save the figure as a **PDF** (and optionally also PNG).

```
using CSV, DataFrames, Plots

table = CSV.read("assets/data/ohm_law.csv", DataFrame)
voltage_V = table.voltage_V  # V
current_A = table.current_A  # A

V_bar = sum(voltage_V) / length(voltage_V)  # V
I_bar = sum(current_A) / length(current_A)  # A
dV = voltage_V .- V_bar  # V
dI = current_A .- I_bar  # A
conductance = sum(dV .* dI) / sum(dV .^ 2)  # S
intercept = I_bar - conductance * V_bar     # A
fitted = conductance .* voltage_V .+ intercept  # A

p = plot(voltage_V, current_A; seriestype = :scatter, label = "
    measurements",
          xlabel = "Voltage (V)", ylabel = "Current (A)",
          title = "Ohm-law measurements and fitted model", framestyle = :
    box,
          size = (800, 500))
plot!(p, voltage_V, fitted; label = "least-squares model")
savefig(p, "ohm_law.pdf")   # required
savefig(p, "ohm_law.png")   # optional
```

What to submit

Submit this bonus on the **Bonus 1 submission box** in Moodle, hand-graded. Include:

- **The figure as a PDF** (the required output). A PNG may be added, but the PDF is the one that is required.
- **Your program**, either uploaded as a file named `ohm_law.jl` or pasted as text in the submission box.
- **One sentence** giving the fitted conductance and whether the data pass the $V = 0 \Rightarrow I = 0$ check.

Do not submit a screenshot of the figure: the PDF must be the saved plot file. There is no CodeRunner question for this bonus; a human reads your code and figure.

Checklist used for grading

Prescribed title; axis labels with units; a `measurements` marker legend and a `least-squares model` line legend; boxed frame; canvas $(800, 500)$; a correct centered-sum fit; the code included as a file or text; and one interpretation sentence.